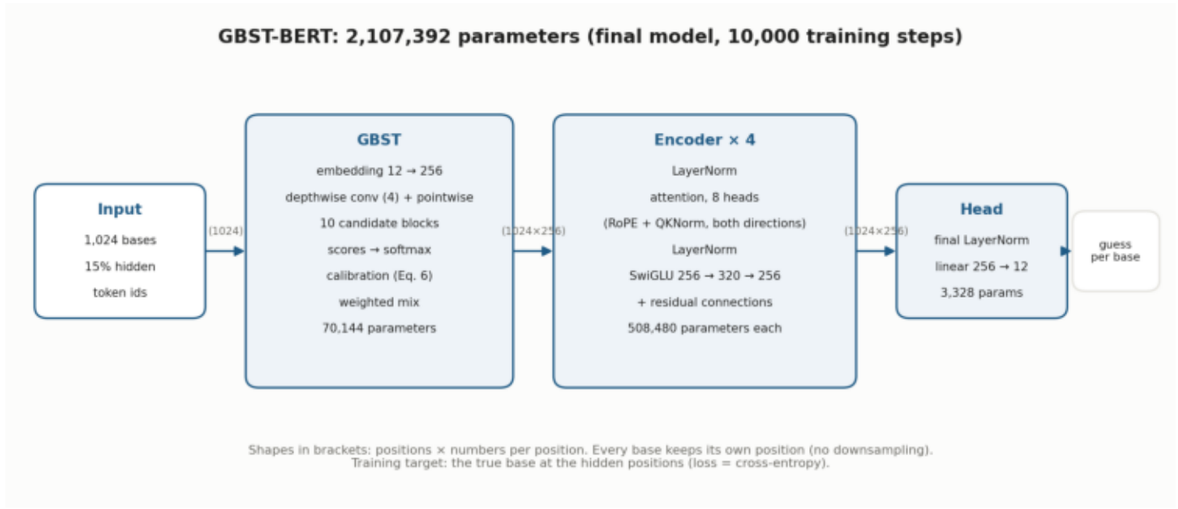
Inside GBST-BERT: four views of the final model

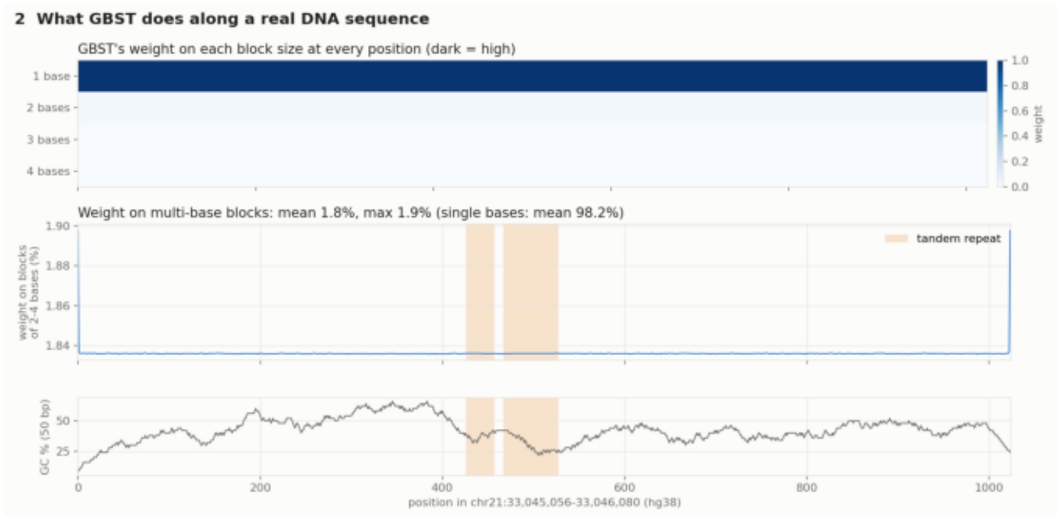
Final model after 10,000 training steps (10.5 billion bases of hg38); example window chr21:33,045,056-33,046,080; tandem repeats marked by rule (unit of 1-6 bases repeated over ≥12 bases).

1 How the model is built



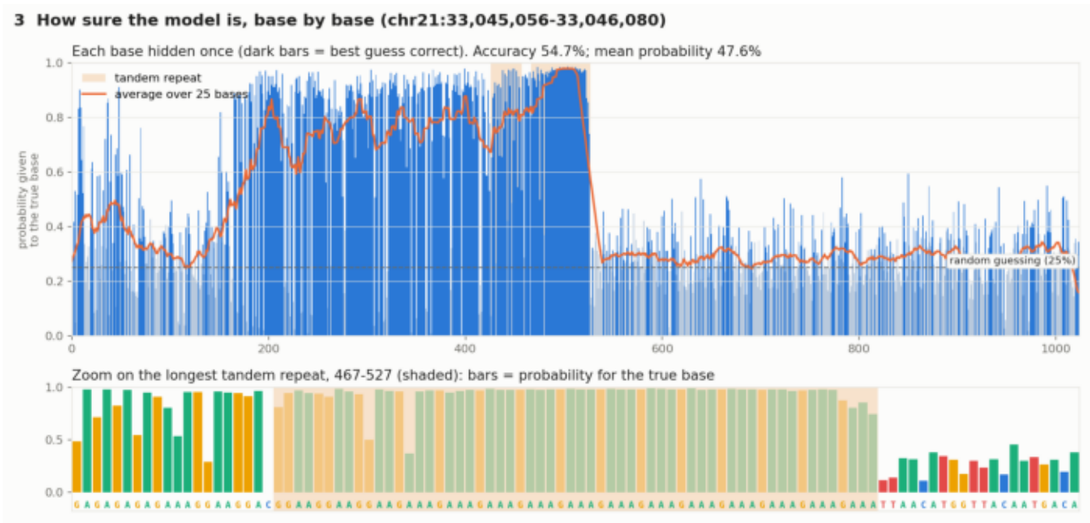
DNA (1,024 bases, 15% hidden) → GBST (learned reading, 70,144 parameters) → 4 encoder blocks (attention looks both ways, 508,480 parameters each) → a guess for every base. Total 2,107,392 parameters.

2 What GBST does along a real DNA sequence



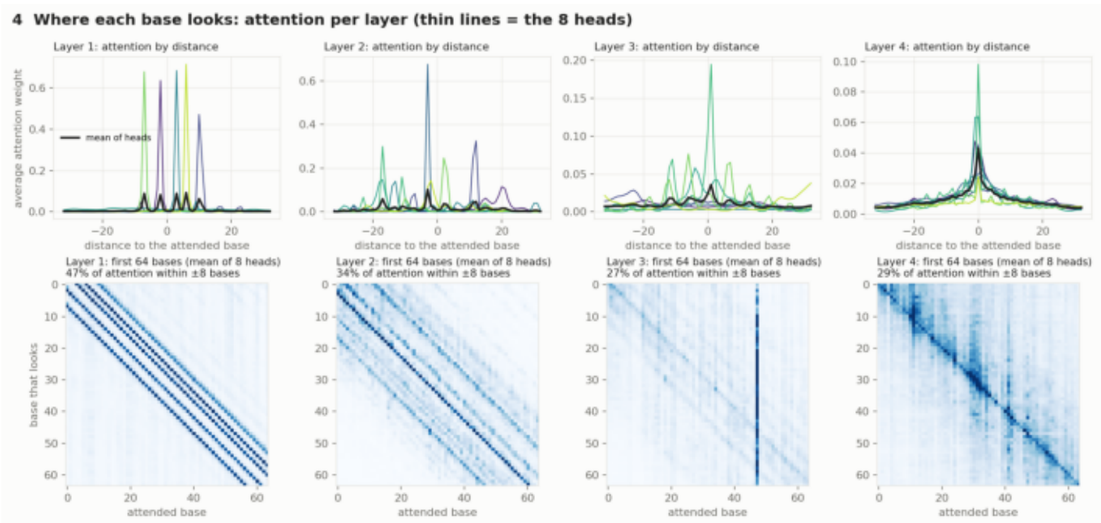
At every position GBST puts 98.2% of its weight on single bases. The small weight on 2-4-base blocks is the same inside and outside repeats (shaded) and does not follow GC content: GBST does not adapt.

3 How sure the model is, base by base



Each base hidden once. Inside the tandem repeat (shaded) 94.5% of bases are guessed correctly; elsewhere 50.8%. The model has learned that repeats are predictable (54.7% in this window).

4 Where each base looks (attention)



Layer 1: every head looks at one fixed nearby position (a few bases left or right). Deeper layers look further (47% → 29% of attention within ±8 bases). Layer 3: one position is looked at by all bases.

Main message: the model reads local context through the convolution and attention, not through GBST's block choice.